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RESEARCH ARTICLE

Blockchain Integration With Multimodal Biometric Authentication System for Secure Blockchain Voting with Face Authentication

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ABSTRACT The prevalence of political interference during election processes remains a significant challenge to the free, fair, and credible conduct of general elections. This election event is crucial as a pillar of democratic governance in any democracy, with the potential for unfitting disturbance and chaos, such as multiple votes, illicit voting, and malicious actors that can be exploited electronically to disrupt voting or affect vote counts. However, this research proposed to develop a secure, smart, verifiable e-voting system (SSVEVS) that can offer an authenticated end-to-end tally voting system, a top-secret ballot election system, and confidence in overall election integrity. The proof of this smart and secure electronic voting system utilizes a 64-bit quad-core ARM Cortex-A76 processor, integrated with multimodal biometric (facial and fingerprint) authentication systems, programmed with a deep-learning image processing (DLIP) algorithm to optimize image detection and recognition. Also, an Ethereum blockchain technology (EBT) with a homomorphic encryption algorithm was implemented on the system to ensure that the original information record is maintained, immutable, tamper-resistant, and transparent throughout the electoral process, and stores the information in a decentralized database application. The system achieved 1,424.501 TPS for 10,000 transactions with a mining time of 7.02 seconds. The biometric (facial and fingerprint) authentication achieved a False Acceptance Rate (FAR) and False Rejection Rate (FRR) of 0.00% respectively. The True Acceptance Rate (TAR) is 100%, with a false image template of 0.04% true identification rate (TPIR).

INDEX TERMS ARM Cortex-A76, biometric, blockchain, deep learning, electronic voting, fingerprint, image processing.

I. INTRODUCTION

The concept of voting in Ancient Greece and Rome was designed for leadership selection in a democratic decision-making process [1]. The reform of democratic politics and policies was the mission of the Intergovernmental Institute for Democracy and Electoral Assistance (IIDEA) in Africa, Asia,

Latin America, and Sweden [2]. This assists in comparative knowledge that supports sustainable changes in democracy. The most challenging part of the electioneering process in a democracy is upgrading the voting process technology. This upgrade in technology can reduce and control human influence in the casting and vote counting through the computerized voting system [3]. The statistical analysis shows that 21% of countries have fully implemented the e-voting system for their democratic decision-making, 57% have never

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implemented it, and 6% have abandoned use of e-voting [3], [4]. The feasibility studies show that some countries practicing democracy without implementing an e-voting system are 16% [4].

The conventional voting in Ancient Greece and Rome was classified into five categories. This includes ballot papers and box-based manual counting systems, mechanical-based voting machines, punch card ballot systems, optical scanner-based ballot systems, and direct recording-based electronic voting systems [5]. Each of these categories of voting methods has some deficiency, such as count bias and result postponement or delay, an unavailable backup for the votes counted, and the impossibility of recounting votes. Lack of an intuitive system to decipher the votes cast from a mispronounced card system and inappropriate marking of the ballots resulted in voter impersonation, miscalculation, and biased result declaration [6].

The electronic voting system is regarded as the most effective and efficient method for casting, counting, storing, and securing votes against impersonation. It replaces traditional voting methods and protects voters' confidential information, ensuring "one person" equals "one vote" without compromising data privacy [7]. E-voting is an advanced democratic tool designed to build trust in electoral management, enhance the credibility of election results, and improve the overall efficiency of the voting process. It also aims to provide a faster election process, eliminate common fraud risks, increase accessibility, and make voting more convenient for citizens [8].

However, these online and electronic voting systems face more criticism than any other information technology (IT) application in election activities, privacy breaches, and impersonation in democratic decision-making processes [9]. This is due to a low-tech spec, insufficient technical materials, poor governance, and oligarchic tendencies [10]. The chaos of this election fraud, multiple voters, and lack of trust in the verifiability of authentic users contribute to the challenges of electoral misconduct and its credibility [11], [12], [13].

The antecedents of electoral malpractices and anomalies resulted in;

- i. A lack of free, fair, and credible elections for leadership selection
- ii. Non-coherent voter registration
- iii. Voter impersonation
- iv. Insecure centralized database of record-keeping in the e-voting system
- v. Ballot filling with bribery
- vi. Counting errors
- vii. Distribution of election materials from central locations to polling centers, and
- viii. External interference by polling agents [14], [15].

These malpractices cost the country billions of dollars, hundreds of lives, property destruction, and a big drag on the nation's economy [16]. For instance, the Nigerian population was reported to be 200,963,599, with a median age of 17.9 years. Approximately 100 million Nigerians were

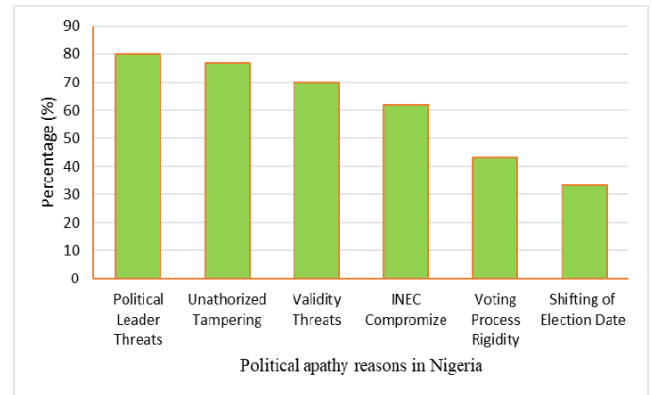


FIGURE 1. Reasons for Nigerian political indifference.

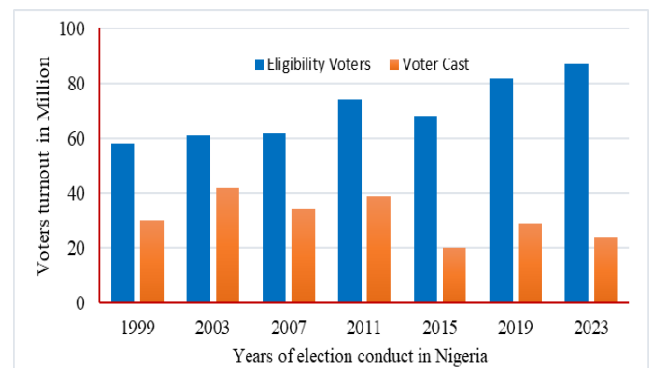


FIGURE 2. Statistical analysis of voters' turnout in the past general elections.

eligible to vote; the statistical analysis shows 28,614,190 voters participated in the election, which is far lower than the potential Nigerian voter population in the election exercises [17], [18]. Political apathy, malpractices, threats, and insecurity contribute to the low participation. It was observed that the number of voters participating in the general election between 1999 to 2023 was drastically reduced due to some reasons identified in Figure 1. Figure 2 shows the differences.

Therefore, the investigation of electoral fraud through the suspected activities is the most potent threat to democratic transition and the sustainability of democracy. These various categories of electoral fraud globally seem to have no end [19]. Also, the failure of democracy to deliver development has contributed immensely to making many Nigerians lose interest in election activities, and violence and voter suppression also played a huge role in diminishing voter turnout [20]. This sabotage and threat require maximum attention from the government and collaboration for adequate coordination, planning, monitoring, and effective control to achieve an extreme level of free, fair, and credible electoral processes.

A. BIOMETRIC SURVEILLANCE AND ITS USABILITY

Biometric authentication is a unique physiological and behavioral feature that can be used for individual identity verification and authenticity. This biometric technology has been

widely adopted in e-voting systems to verify the authenticity and prevent voter impersonation [18]. It increased security, transparency, voter confidence, and reduced electoral fraud. Some biometric traits frequently used in internet-based electronic voting systems for electorate registration, verification, and authentication are facial, fingerprint, iris, retina, palmprint, hand geometry, and voice recognition [17], [21].

However, the use of biometric surveillance in enhancing the e-voting system introduces issues in handling sensitive biometric data, accessibility for voters with disabilities, and data privacy breaches [22]. Another issue associated with this biometric-based e-voting system is the technical infrastructure related to hardware and software, unbiased, and accuracy. The user acceptability of the e-voting system and its scalability issues are influenced by trust, usability, accessibility, and perception of security [23]. The analysis of adopting biometrics in an e-voting authentication system was examined based on acceptability, uniqueness, permanence, measurability, and universality.

- i. Acceptability: refers to the level of support for the biometric feature
- ii. Permanence: implies that the biometric feature does not vary and is stable over time
- iii. Uniqueness: refers to the biometric feature peculiarity of physical and behavioral exception characteristics
- iv. Measurability: implies that the biometric feature can be practically collected and analyzed for unique characteristics and verifiable
- v. Performance: refers to the level of invariance of the biometric feature over time
- vi. Universality: implies that the biometric feature that occurs in every human is a quality (being true) and is appropriate all the time.

The usability analysis of the multifactor biometric authentication and verification system is conducted and presented in Table 1. The sum of usability $\sum U_{\theta}$ for each biometric feature in the developed e-voting system. The numeric values assigned to the usability ratings are L=5=Low, M=10=Medium, and H=20=High. The high performance and universality offered by the DNA biometric trait are unsuitable for authentication in e-voting, as it has low acceptability and measurability, but has highly intrusive and expensive characteristics.

where, BF is the Biometric Feature, U_q is the Uniqueness, P_m is the Permanence, U_v is the Universality, P_f is the Performance, A is the Acceptability, M is Measurability.

B. BLOCKCHAIN-BASED E-VOTING SYSTEM AND ITS SECURITY

Blockchain technology improves the security level in both online and electronic-based voting systems and biometric authentication [24]. This offer advantages such as immutability, transparency, traceability, authentication, privacy, trust, data confidentiality, and integrity [25], [26], [27], [28]. Blockchain is a distributed ledger that stores data through a mining algorithm and consensus mechanism [29], [30].

TABLE 1. Biometric features and usability analysis.

BF	U_q	P_m	U_v	P_f	A	M	$\sum U_{\theta}$
Facial	5	10	20	5	20	10	70
Hand Geometry	10	10	10	10	10	20	70
Retina	20	10	20	20	5	5	80
Voice	10	5	10	5	20	5	55
DNA	20	20	20	20	5	5	90
Signature	5	5	5	5	20	10	50
Hand Vein	10	10	10	10	10	5	55
Fingerprint	20	20	10	10	20	20	100
Iris	20	20	20	20	10	20	110

where, BF is the Biometric Feature, U_q is the Uniqueness, P_m is the Permanence, U_v is the Universality, P_f is the Performance, A is the Acceptability, M is Measurability.

The blockchain-based smart contract allows all nodes in the distributed ledger to reach a consensus on the hashed blocks [31]. This transaction is mined before being broadcast to every node connected to the peer-to-peer network for adequate security and verification. The blockchain principle of operation is illustrated in Figure 3.

A privacy-preserving distributed voting system is developed for human voting and voting activities [32], [33]. A distributed electronic voting system offers secure and transparent parliamentary election activities in countries such as Australia, Estonia, and Norway. Using a modified Proof-of-Stake (MPoS) consensus algorithm to achieve a better performance system [34].

The e-voting system, powered by blockchain technology, offers the advantages of transparency, non-repudiation, decentralization, and irreversibility in the voting mechanism. This system provides a feasible general election and meets the requirements [35]. A conceptual framework for a secure electronic voting system using blockchain technology is developed to accomplish reliable, anonymous, and tamper-resistant voting information [36].

The quest to achieve confidentiality, integrity, transparency, and privacy in e-voting systems has focused on adopting a crypto-hashed decentralized system called blockchain technology, which offers the immutability of votes, voting transparency, decentralization of authority, and confidentiality in vote counting [29]. This secure communication and exchange of information between two parties uses cryptography, mining, and a consensus algorithm to prevent third-party interception. A summary analysis of existing works that use blockchain technology to enhance security, biometrics for voter authentication, verification, and integrity is presented in Table 2.

C. CONTRIBUTIONS TO KNOWLEDGE AND ORGANIZATION OF RESEARCH

This research develops a secure, smart, and verifiable e-voting-based blockchain and multi-factor biometric authentication systems, using a facial and fingerprint sensing system. These combined biometric usability factors of blockchain technology will ensure secure end-to-end

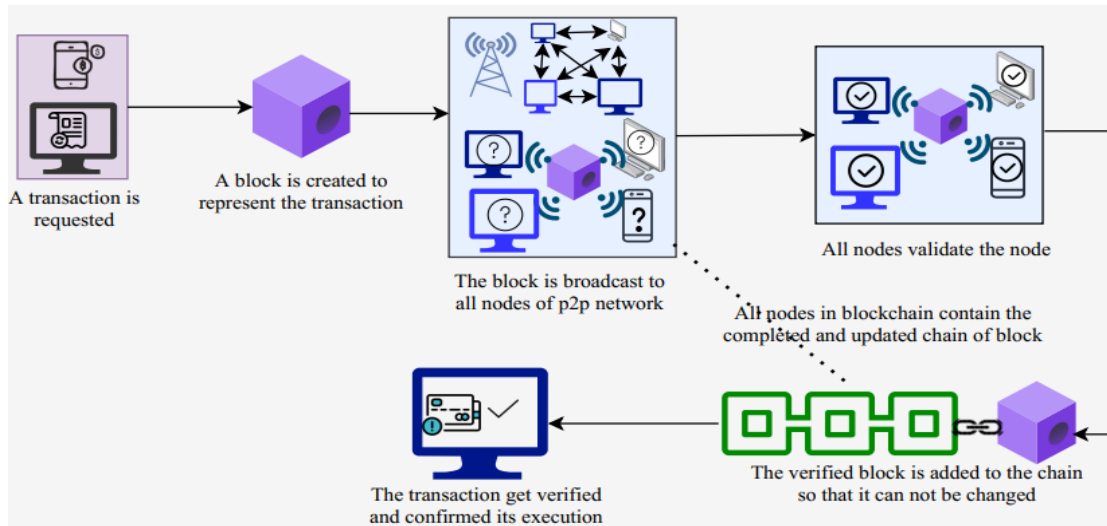


FIGURE 3. Blockchain transaction overview [44].

TABLE 2. Analysis of blockchain, biometrics, and controller's applications in the voting systems.

Voting system		Controller		Security mechanism				Biometric device		Reference
E-voting	O-voting	Microchip	Blockchain	Voter ID	OTP	Smart card	Voting key	Fingerprint	Facial	
Yes	--	--	Yes	Yes	--	--	--	Yes	Yes	[1]
Yes	--	--	Yes	--	--	--	--	Yes	Yes	[5]
Yes	--	Yes	Yes	Yes	--	--	--	Yes	Yes	[7]
--	Yes	--	Yes	Yes	--	--	Yes	--	--	[9]
Yes	--	--	Yes	Yes	--	Yes	--	--	--	[10]
Yes	Yes	--	Yes	Yes	--	Yes	Yes	--	--	[11]
Yes	--	--	Yes	Yes	--	Yes	--	--	Yes	[18]
Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	--	[21]
Yes	Yes	--	Yes	Yes	--	--	--	Yes	--	[32]
--	Yes	--	Yes	Yes	--	Yes	Yes	--	--	[36]
Yes	--	Yes	Yes	Yes	--	Yes	--	Yes	Yes	Our work

verifiability and a reliable authentication system. Consequently, the author's contribution to knowledge in this work is highlighted as follows.

i. Develop a secure multi-factor, multi-modal biometrics voting authentication system using a combined facial and fingerprint biometrics sensor integrated with a 64-bit quad-core ARM Cortex-A76 processor

ii. Develop an intelligent deep learning image processing (DLIP) algorithm for multimodal biometric authentication system enhancement using Hough transform, Gabor filter, and Convolution Neural Network (CNN) for image recognition and verification

iii. Develop a secure decentralized database system using Ethereum testnets-based blockchain technology with a Pallier homomorphic encryption algorithm

iv. Performance evaluation of the fabricated system and comparison with the state-of-the-art using accuracy, precision, sensitivity, and speed.

The remaining part of the research is presented as follows. Section II presents related works, and Section III presents materials and methods. It discussed the hardware system

design of SSVEVS, the Raspberry Pi module, Raspberry Pi Camera module, fingerprint module, and RFID module, which were presented in 3.1. The image processing algorithms for biometric sensors and a convolutional neural network (CNN) for image classification were presented in 3.2. The Ethereum blockchain technology for securing the e-voting system is discussed in 3.3. Results and discussion were presented, testing and validation of SSVESS, and evaluation of the Ethereum blockchain-based e-voting system. Section V presents the conclusion and recommendations for future research.

II. RELATED WORKS

Recently, a semi-automated paper ballot machine has been used to address the challenges of the previous paper ballot system. Still, it does not implement adequate security measures during voter registration and validation due to technical and socio-technical vulnerabilities of the system [16]. Smart Card Readers (SCRs) based voting in the 2015 and 2019 general elections played a significant role in securing centralized records [37], [38]. The Bimodal Voters

Authentication System (BVAS) by the Independent National Electoral Commission (INEC) faced several challenges, like privacy breaches, record tampering, and result alteration [39], [40]. These technical imperfections of BVAS bring some issues of dissatisfaction in Nigeria's general election process.

The online voting-based blockchain technology was developed to increase voter turnout and reduce organizational costs [41], [42]. This approach eliminates the use of a ballot paper system and increases the election process. Introduces malicious hackers and social threats, a single vulnerability that can result in vote rigging and manipulations. An electronic voting-based blockchain was developed to authenticate users through a fingerprint recording system and the lightweight block cipher Speck [34]. Voter security, authentication, and data integrity are all guaranteed.

Salman et al. proposed a secure Blockchain voting mechanism that guarantees both device and node security, a safe and transparent voting procedure using elliptic curve Cryptography (ECC) to encrypt data on the blockchain. The system was simulated using NS3 with a predefined number of IoT devices or nodes [43]. The proposed system is secure, transparent, prevents double voting, is consistent in vote counting, and can detect harmful nodes on the network. However, as the number of nodes increases, the system tends to slow down, making the IoT nodes vulnerable to attacks.

Salman. A Double Blockchain-based E-voting (DBE-voting) system using private-public blockchain technology on Hyperledger Fabric [44]. The voter information is recorded on the private blockchain, and the voting results are recorded in private and public blockchains. The private blockchain helps protect voters' privacy, and the voting results can be cross-checked in the public blockchain to ensure election integrity. The hybrid private-public on-chain and off-chain storage was adopted to ensure the consistency and correctness of voting data on the platform. The facial ID and fingerprint user identity are used before authorizing an electorate. The system fulfills five core requirements in E-voting: authentication, correctness, privacy, and reusability. However, the system has issues of storage redundancy, data consistency between the two storage systems, and low throughput.

Liu et al. proposed a secure Blockchain-based E-Voting system using Speck cipher, a lightweight block cipher, and baker's chaotic key generator to secure voters' information on each block in the blockchain [45]. A fingerprint logging system authorized voters' eligibility to vote using Scale-Invariant Feature Transform (SIFT) and K-Nearest Neighbors (KNN). The e-voting system allows voters to choose anywhere with internet access. The system consumes less energy, has a low response time, and has high accuracy during authentication. However, it is less secure than other blockchain encryption algorithms and is susceptible to attacks because of the risk of logging in from any network.

Mohsen et al. proposed a secure online-voting system (SOVS) blockchain technology for data security, a registration card, and facial ID for verification and authentication [46]. The SOVS was enhanced and further

strengthened the security measure through email verification for one-time password (OTP) confirmation. The blockchain was implemented on the Solidity platform to prevent double voting. However, the system's authentication accuracy is weak and prone to fraud, but it has high integrity.

Preiya et al. proposed an e-voting technology with evolutionary theory (Genetic Algorithm). This system incorporates digital signatures, Homomorphic Encryption (HE), zero-knowledge proofs (ZKPs), and Byzantine fault-tolerant consensus techniques to ensure security while maintaining transparency and voter anonymity [47]. It employs a permissioned blockchain featuring a hybrid consensus mechanism that combines Proof of Work (PoW) and Proof of Authority (PoA). Built on Docker, the system is deployed using Kubernetes. Overall, it offers strong security while being cost-effective, transparent, tamper-resistant, and ensuring anonymity. However, it faces challenges with scalability and requires considerable computing resources.

Pereira et al. proposed an e-voting Privacy and Verifiability Preserving Blockchain (PVPBC). The system uses a fully verifiable e-voting protocol to publish votes in plaintext accompanied by tracking numbers, which enables voters to confirm that their votes have been counted by the system [48]. This system was built on the Quorum blockchain network, a permissioned ledger and smart contract. A voter's ID was used as a means of voter authentication. The system supports voter privacy, anonymity, ballot privacy, security, scalability, and fairness. However, the system is slow, and the gas is high during operation.

Faruk et al. proposed a blockchain-based e-voting system deployed on the Linux platform [49]. The system helps improve the security of the votes, eliminates the problem of vote falsification, makes the voting process efficient, reduces the cost of election security, and maintains voter privacy. However, the voting process is too complex for voters, and there is no authentication procedure to remove ineligible voters or prevent double voting. Conclusively, other related works on the developed e-voting system, various techniques, and security of the voting records using biometrics, cryptography, and blockchain are presented in Table 3.

III. MATERIALS AND METHODS

The hardware design of a secure, smart, and verifiable e-voting system (SSVEVS), and software system programming design based on a secure decentralized blockchain, as presented in Figure 4. The hardware-based system uses a 64-bit quad-core ARM Cortex-A76 processor as a central controller, a wireless radio frequency identification (RFID), and a bimodal authentication system (facial and fingerprint sensors). This 64-bit quad-core processor stores bimodal authentication images, programmed with a deep-learning convolutional neural network (DLCNN). The histogram equalization and the Gabor filter enhance the templates (fingerprint and facial images). The Hough transform and Gabor filter are used to improve the sharpness of the image's feature extraction.

TABLE 3. Summary of the related work analysis based on secure mechanisms, methods, strengths, and limitations.

Title	Methods	Strengths	Limitations
Investigating Performance Constraints for Blockchain-Based Secure E-Voting System [50].	Experimental setup: Permissioned (Multichain) and permissionless blockchain environments. Parameters Tested: Block size (1 MB to 1 GB), block generation rate (15–60S), transaction speed, voter population (10–222,000). Tools: Multichain platform, Java-based clients, JSON-RPC API.	Rigorous scalability analysis for real-world e-voting applications. Identified trade-offs between block size, generation rate, and security. Demonstrated feasibility for large-scale elections (e.g., 222,000 transactions with 7 concurrent clients).	Limited to the Multichain platform; results may not generalize to Ethereum or Bitcoin. Network latency simulations may not fully replicate real-world conditions. Focused on e-voting; applicability to other domains (e.g., finance) requires further study.
Self-tallying E-voting with Public Traceability Based on Blockchain [51].	Cryptographic techniques: Time-lock puzzle, traceable ring signature, ZKPoK (Zero-Knowledge Proof of Knowledge). Consortium blockchain implementation (Hyperledger Fabric). On-chain/off-chain hybrid architecture with smart contracts.	Solves adaptive and abortive fairness issues via time-lock puzzles. Public traceability detects double-voting without a central authority. Balances voter anonymity with accountability. Implemented and tested for practical performance (e.g., 244ms per ballot cast).	High computational overhead for cryptographic operations (e.g., ZKPoK). Scalability concerns with large voter populations due to linear growth in verification time. Reliance on strong cryptographic assumptions (e.g., DDH, sequential squaring).
SecureBallot: A secure open-source e-voting system [52].	Cryptographic techniques (AES, RSA, HMAC) for encryption and authentication. Token-based voter identification using NFC tags. Formal protocol verification via Casper/FDR. Case study in university elections.	Ensures voter anonymity and integrity. Open-source design enhances transparency and trust. Formal verification confirms security properties. Tested in real-world scenarios with high user satisfaction.	Requires supervised physical polling stations and convenience. Dependency on NFC infrastructure may pose logistical challenges. National elections have not been fully validated for scalability.
A biometrics-generated private/public key cryptography for a blockchain-based e-voting system [53].	Blockchain for immutable vote storage. Biometric fingerprint data for key generation. ECDSA for public key derivation. Smart contracts for vote management. Preprocessing (Otsu thresholding, skeletonization) for fingerprint feature extraction. SOCOFing dataset for testing. MetaMask integration for DApp connectivity.	Biometrics ensure uniqueness and resistance to forgery. Blockchain guarantees transparency and immutability. Voter anonymity via public keys. Tested with 400 voters, demonstrating practical feasibility. Seamless authentication via fingerprint integration.	Requires fingerprint hardware, limiting accessibility. Computationally intensive preprocessing steps. Scalability for large elections (e.g., national) has not been fully validated. Blockchain latency in high-volume transactions.
DVTChain: A blockchain-based decentralized mechanism to ensure the security of a digital voting system [54].	Blockchain Platform: Ethereum 2.0 with Solidity for smart contracts. Key Components: Voter/Candidate/Voting smart contracts, cryptographic server for key management. Encryption: Votes are encrypted using public keys until the election ends. Verification: Merkle trees for data integrity, vote coins for uniqueness. Tools: Truffle, Ganache, MetaMask for deployment.	Ensures anonymity (voter info stored as hashes). Fairness (votes encrypted until election closure). Verifiability (individual and universal). Reduced transaction costs compared to existing systems. Mobility (accessible via internet-enabled devices).	No OTP implementation for registration. High storage costs due to encrypted votes on-chain. Scalability challenges for large-scale elections. Reliance on internet access and digital literacy.
Blockchain Ethereum Clients Performance Analysis Considering E-Voting Application [55].	Blockchain Platform: Ethereum network with Geth and Parity clients. Key Components: Smart contracts for e-voting (owned, token recipient, election), Remix IDE, Web3.js, EVM. Metrics: Throughput, latency, scalability. Experimental Setup: 10-node cluster with Ubuntu 16.04, 1.5GHz CPU, 8GB RAM.	91% faster transaction processing in Parity vs. Geth. Ensures fairness, eligibility, privacy, and verifiability for e-voting. Practical scalability analysis (tested up to 10 nodes).	Limited scalability testing (only 10 nodes). No security analysis against attacks (e.g., Sybil, DDoS). Focus on private blockchain, lacking insights into public blockchain challenges. No discussion of energy consumption or user experience.

The software system is a secure, decentralized database for voter registration management, authentication, vote counting, and validation. This ensures that the original information record is immutable, tamper-resistant, and transparent throughout the election process. The mode of operation of this smart, verifiable e-voting system, with electorate registration through the electorate authority (EA), ballot transaction, encryption, decryption, vote transaction, vote tally, and broadcast of results, as presented in Figure 5.

This SSVEV allows the electoral authority (EA) to register the electorate E with personal biodata, and capture the fingerprint and facial data, which are all encrypted with PHC and hashed before being added to the blockchain network using a proof-of-stake consensus algorithm. The token (gas fees) is created and added to the wallet of individual electorates' accounts as a unique authentication for the voters' V. During the voting process, the voter's biodata information will be verified in addition to the biometric authentication record using an RFID tag to prove eligibility to vote. Immediately

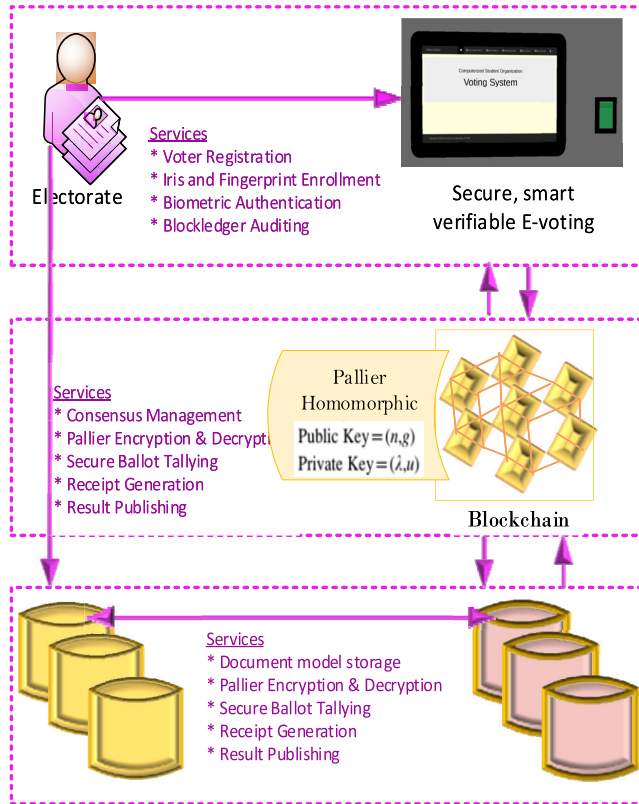


FIGURE 4. The workflow of the secure, smart, and verifiable e-voting system-based blockchain architecture.

after verification and authentication, a token will be assigned to the wallet with a minimum amount that may be enough for the voting positions. After voting, secure information will be sent to the electorate's email account, along with a private access key to ensure the vote is counted, verified, and auditable.

A. HARDWARE SYSTEM DESIGN FOR SECURE, SMART, AND VERIFIABLE E-VOTING SYSTEM (SSVES)

The hardware system design for secure smart e-voting is implemented on a Raspberry Pi 4 controller system, embedded with an intelligent Single-on-Chip (SoC) ARM Cortex-A72 processor for processing and connecting electronic applications. This adaptable, low-cost small computer has general-purpose input/output (GPIO) pins for connecting peripherals. The connected Pi camera module captures facial images through a standard Camera Serial Interface (CSI), and the host processor stores the frame in memory via a virtual Mobile Industry Processor Interface (MIPI). The fingerprint sensor module (FPM10A) is connected to the SoC processor through a universal asynchronous receiver and transmitter (UART) serial port for image template acquisition. This FPM10A sensor is designed for fingerprint enrollment/verification, reading/writing fingerprint template files, and extracting fingerprint images using Frustrated Total Internal Reflection (FTIR) protocols.

The biometric images acquired from these devices are optimized and enhanced for sharpness using histogram

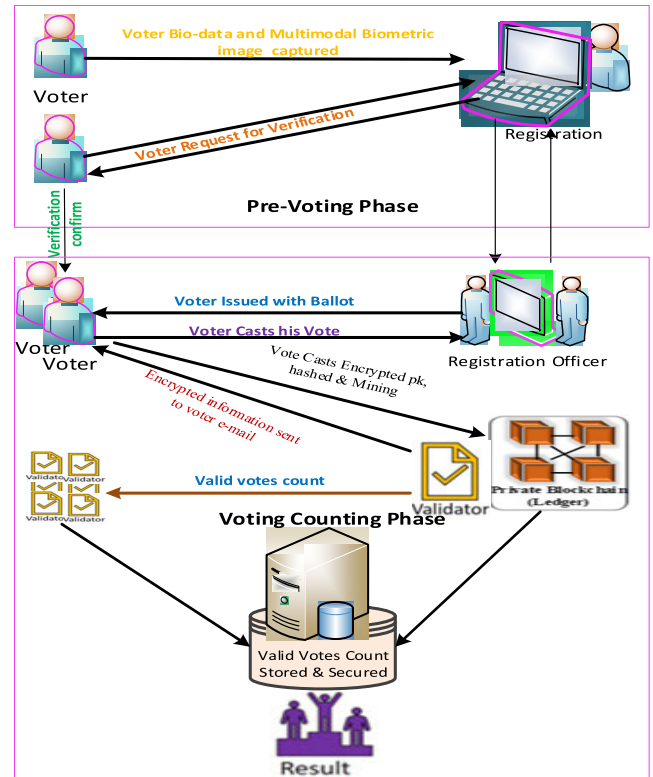


FIGURE 5. Architecture of a smart, secure e-voting system.

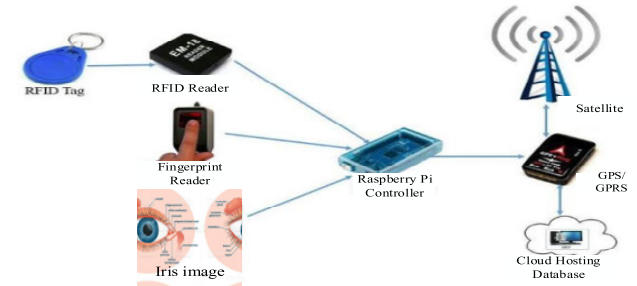


FIGURE 6. Multimodal biometric authentication system architecture.

equalization, Hough transform image processing, the Garbor filter for feature extraction, and filtering. Then, a deep convolutional neural network (DCNN) is developed for enrollment image templates, recognition, and matching through a Python interactive programming environment, using the SD flash memory. The radio frequency identification (RFID) utilizes electromagnetic fields and is integrated with compatible RFID tags on each electorate, automatically identifying the authentic user with unique electronically stored information. This RFID device is interfaced with a Raspberry Pi through GPIO pins. The integrated hardware system overview of the SSIVes structure is presented in Figure 6, and a flowchart of the processes is presented in Figure 7.

1) RASPBERRY PI

The Raspberry Pi is a single-board mini-computers that create easy access to computing, automation, embedded

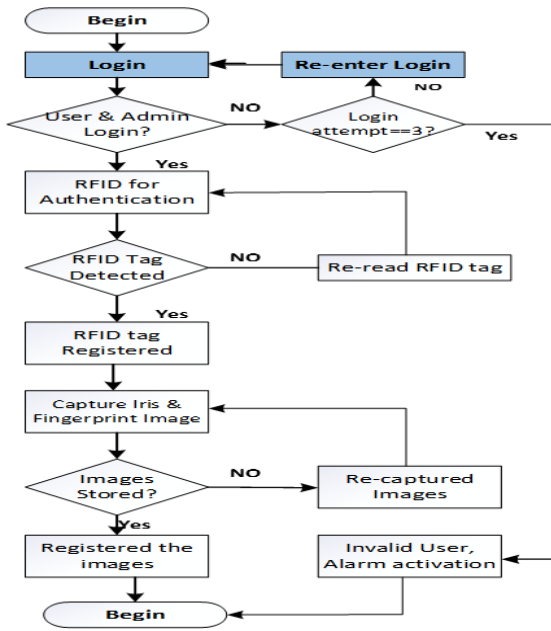


FIGURE 7. Multimodal biometric authentication principle of operation.

systems, programming, and industrial applications. It utilizes a single-core processor of about 700MHz-1.5GHz for data processing and 256MB-4GB RAM for storage. The Raspberry Pi is designed with an integrated ARM Cortex A76 processor that is capable of running both Windows operating system (OS) and Linux, and provides a set of general-purpose input/output (GPIO) pins for the control of physical computing, electronic devices, and for the usefulness in Internet of Things (IoT) projects.

The Raspberry Pi is a high-performance computing system with a low-cost resource platform for education and learning about programming and automation systems. This system is responsible for storing biometric data information and providing security support for the decentralized database of electoral activities. The intelligent image processing algorithms, the biometric image templates (Fingerprint and Facial), and Blockchain integration were stored on the 32 GB secure digital flash memory card through an SD card slot on the Raspberry Pi.

However, several considerations were involved when choosing the Raspberry Pi-based ARM Cortex-A76 processor as the controller unit for this SSIEVE system. The ARM Cortex-A76 processor performs excellently when an appropriate power output of 5VDC/5A is provided through a buck-boost converter design, as shown in Figure 8. The buck-boost converter DC-DC power supply has a 5VDC/5A/25W output voltage. This ensures a stable power supply, over-current protection, and an optimal Raspberry Pi-based e-voting system. The Raspberry Pi's efficiency depends on the utilization of power requirements, which increase based on the number of input/output interfaces connected.

A comparison of relevant features among several microchip-based development boards, system performance,

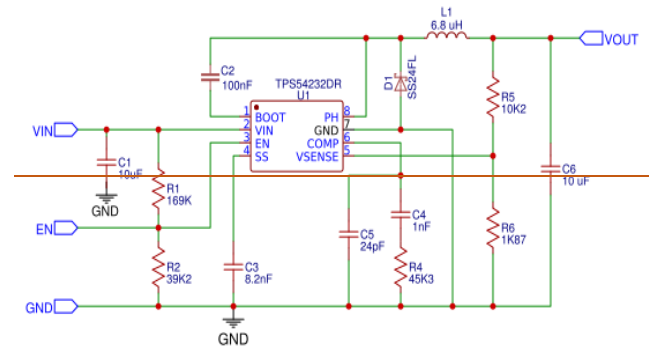


FIGURE 8. Circuit design for 5V/60A buck-boost power supply.

onboard wireless connectivity, power consumption, size, and weight. The Raspberry Pi and Intel Edison have better industrial applicability, reliability, processor speed, and memory size. But excessive heat generated and power consumption necessitate the provision of an internal cooling fan as presented in Table 4.

2) FINGERPRINT SENSOR MODULE (FPM10A)

This sensor module captures the electorate's fingerprint identity (ID) during user identification and authorization. The FPM10A fingerprint sensor module has flash memory that stores 127 different fingerprint templates. It has a false accept rate (FAR) of less than 0.0001% and a false reject rate of <1.0%. The module can be powered through a 3.6-6.0 DC voltage supply and 120mA current. This sensor comprises six pins: Don't connect, VCC, TX, RX, and GND. The digital pins TX and RX of the sensor module are connected to the GPIO pins 14 & 15 of the Raspberry Pi, VCC and GND are connected to pins 4 & 6 of the board, respectively.

3) RASPBERRY PI CAMERA-ASSISTED IMAGE CAPTURING MODULE

This is a facial recognition camera integrated for capturing and processing through the system's graphics processing unit (GPU). The camera module is optimized with algorithms such as histogram equalization, Hough transform image processing, Gabor filter, and CNN for face detection, feature extraction, image filtering, and image sharpness acquisition, as well as training for recognition. The input frame is captured by the Raspberry Pi camera and then processed for detection using computer vision algorithms. The embedding faces are subjected to the deep neural network for feature extraction and conversion into vector arrays. The vector array stored in the database was trained and compared for face recognition.

4) WIRELESS RADIO FREQUENCY IDENTIFICATION (RFID) MODULE

This wireless device utilizes radio waves for object identification and reading information recorded on passive RFID at a distance of 0.9 meters. The passive ultrahigh-frequency read information from 3-6 meters, while the super high-frequency

TABLE 4. Comparison of various microchip development board features.

Microchip Board	Raspberry Pi	ATMEGA 328	Intel Edison	ESP 32	STM32F205
CPU Speed	1.5 GHz-1.8GHz	16 MHz-20MHz	500 MHz	240 MHz	120 MHz
ROM Capacity	>32 GB	1 KB	>4 GB	>448 KB	1 MB
RAM Capacity	>8 GB	2 KB	>1 GB	>520 KB	128+4KB
Input Voltage	3.5-5V	7-12 V	3.3-4.5V	5V	48.26V
Input Current	1.2-5A	150/1.5mA	-	30-68/0.015mA	1.8-3.6mA
Clock Frequency	300-400MHz	16MHz	12MHz	80-240MHZ	120MHz
Wireless Board	Ethernet, WiFi & Bluetooth	None	Wi-Fi	Wi-Fi & Bluetooth	Wi-Fi & Bluetooth
CPU Features	Quad Core	8-bit AVR	Atom-2 Core	32-bit RISC Single Core	32-bit Cortex M3
GPIOs	28	20	40	30-38	16
Price	\$49.27	\$2.14	\$258.98	\$11.19	\$26.88

active RFID tags can access information on the wireless RFID module up to 100 meters. This RFID consists of two components: the tags and the card reader, which can be passive or active. The tag can store information from serial number one to the last available data tag using radio waves to communicate its identity information to nearby card readers. The Card reader is a mobile entity that emits radio waves and receives signals directly from the tags.

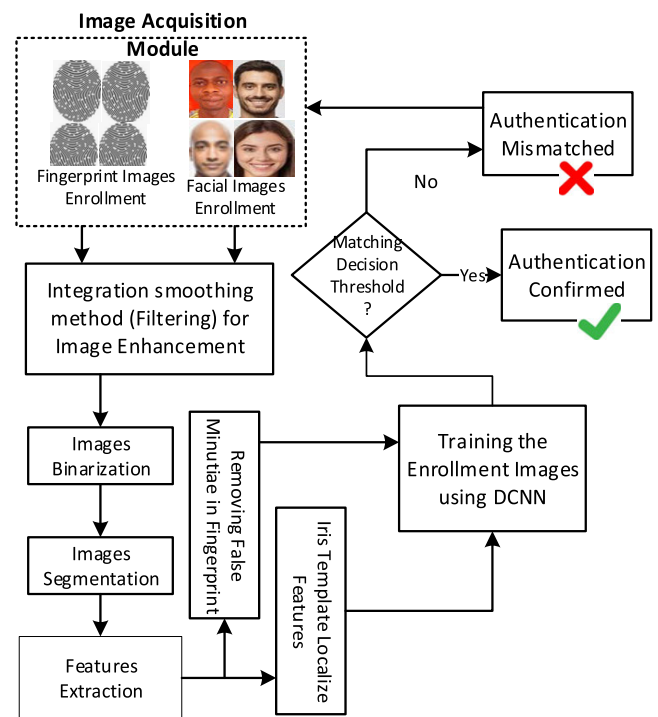
B. IMAGE PROCESSING ALGORITHM FOR BIOMETRIC AUTHENTICATION SYSTEM

The bimodal authentication system uses two or more biometric identification systems to verify the individual voter's identity during the registration stage. And the verification of an electorate participant in the electioneering processes. The fingerprint sensors and facial cameras capture the unique features of the individual candidate for sharpness and fast pattern image recognition. The detailed approach to multimodal image acquisition and processing is presented in Figure 9.

1) AUTHENTICATION ALGORITHM WITH FINGERPRINT IMAGE PROCESSING

Fingerprint biometrics is a unique form of personal identification that involves the flow of ridge patterns in a local region of a fingertip. A fingerprint image typically contains about 40 to 100 minutiae points of interest (PoI), located between ridge bifurcations and ridge endings. However, challenges such as poor image quality and noise can affect recognition, leading to false minutiae detection or loss of genuine features. These issues can be addressed by applying several image processing techniques during pre-processing, such as binarization, segmentation, and histogram equalization. During feature extraction, techniques such as minutiae extraction, image thinning, and singularity detection are used. Finally, false minutiae are eliminated in the post-processing stage.

However, the fingerprint template is preprocessed to adjust the gray value range by reducing the variation that may likely occur along the ridges and valleys. These images are enhanced to improve sharpness by applying median filtering to remove noise and normalizing before introducing

**FIGURE 9.** Block diagram of the bimodal image authentication system.

the histogram equalizer for the template edge enhancement. After the image filtering process, the Gabol filter method is applied with the transformation of the histogram image to acquire a visible black-and-white image of the fingerprint as given in Equation (1), and the feature template generation is achieved as expressed in Equation (2), when, $x_\alpha = x \cos \alpha - y \sin \alpha$, and $y_\alpha = x \sin \alpha + y \cos \alpha$. Biometric fingerprint authentication is presented in Algorithm 1, and Figure 10 illustrates the flowchart processes.

$$h(x, y, \alpha, f) = \exp \left\{ -\frac{1}{2} \left(\frac{x_\alpha^2}{\sigma_x^2} + \frac{y_\alpha^2}{\sigma_y^2} \right) \right\} \cdot \cos(2\pi f x_\alpha) \quad (1)$$

$$\begin{pmatrix} x \\ y \\ \alpha \end{pmatrix} = k \cdot \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & k^{-1} \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ \alpha \end{pmatrix} + \begin{pmatrix} \delta_x \\ \delta_y \\ \alpha \end{pmatrix} \quad (2)$$

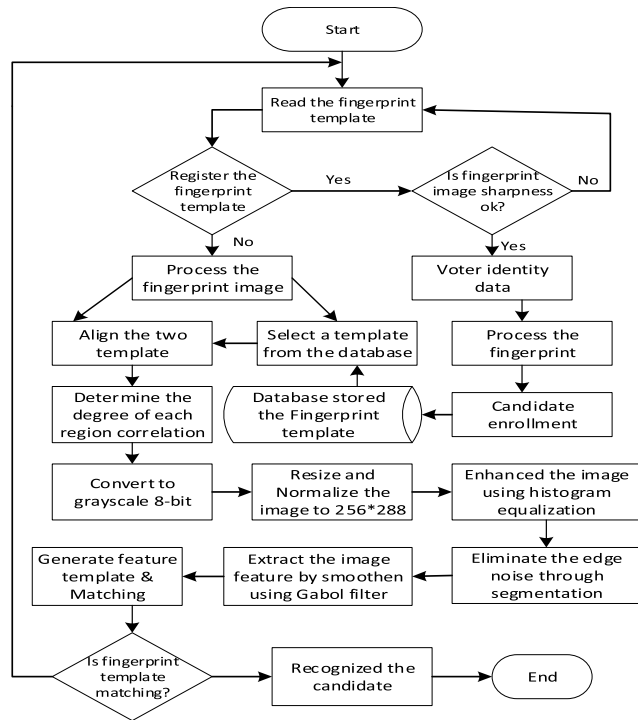


FIGURE 10. Flowchart of the biometric fingerprint authentication system.

where,

- α is the orientation of the fingertip direction
- F is the cosine frequency wave

δ_x, δ_y is the fixed distance from the Gaussian properties along the x-axis and the y-axis

2) HISTOGRAM OF GRADIENT AND HOUGH TRANSFORM FOR FACIAL IMAGE PROCESSING

The facial image is a unique biometric identification and authentication method. It requires an artificial intelligence algorithm for image segmentation and enhancement. This facial image processing and segmentation can be perfectly achieved through various process stages such as acquiring the image, edge detection, description or border location, feature extraction, and post-processing. The facial image phase depends on the electorate image acquisition or enrollment stored in the database for preprocessing and analysis. The preprocessing phase of an electorate image involves transforming from color to gray-scale using a negative, or called inverse, transform to eliminate irrelevant areas. After the transformation of facial image acquisition from color to gray using Equation 3, the binarization and negative image transform are applied to eliminate and reduce the error that may be involved in characterizing the template, and convert the image from gray scale to black and white based on the threshold between 0's and 1's using Equation 4.

The Hough transform image feature extraction and analysis is applied to find the imperfect instances of the image edge point captured within the classes (r, θ) and improve it by

computing the Hough transform to perform estimates of the two parameters' dimensions (x, y) in a free space, as in Equations 5 and 6. Thus, the feature extraction of an iris image is obtained using the Gabor filter algorithm as expressed in Equation 7. The facial image processing for the voters' authentication and verification is presented in Algorithm 2.

$$f_g(x, y) = (R + G + B)/3 \quad (3)$$

$$f'(i, j) = |(2^n) - f(x, y)| \quad (4)$$

$$(x - a)^2 + (y - b)^2 = r^2 \quad (5)$$

such that,

$$\begin{cases} x_1 = r_1 \cos \theta + a_1 \\ y_1 = r_1 \sin \theta + b_1 \end{cases} \quad (6)$$

$$f_{Gabor}(x, y) = \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \cdot \exp(j2\pi f(x \sin \theta + y \cos \theta)) \quad (7)$$

where,

σ Is the filter width of the orientation range coverage ($\sigma = 3.5$)

F The original image, called average spatial ridge-valley frequency ($f = 0.11$)

x, y Represents each pixel along the x-axis and y-axis

a, b Are the coordinates centered?

N Is the number of bits in the image

R Is the circular radius

θ Is the orientation angle between 360°

3) DEEP CONVOLUTIONAL NEURAL NETWORK (DCNN) FOR IMAGE RECOGNITION AND MATCHING

A DCNN is an evolved traditional neural network with multiple hidden layers and uses a 3-dimensional neural convolution to identify image patterns. The DCNN has several layers that provide important functionality to enhance feature extraction of an image pattern, such as input image, convolutional layer, cross-channel normalization, max pooling, rectified linear unit (ReLU), flattening (fully connected), and classification output. The DCNN is adopted in this research to learn the extracted digital biometric images (facial and fingerprint) for classification and segmentation purposes in the secure, smart, and verifiable e-voting system. Bimodal biometric authentication templates of an electorate are represented as an RGB image in the DCNN model for feature learning through the convolutional layer with an integrable function $f * g$ of learnable parameters (weight and bias). This learnable parameter will help to generate the feature mapping as output. The max pooling operation helps to filter the input map features of the biometric images to obtain the optimum value of the filtered region using Equations “(8)” to “(10)”.

$$\gamma_k = f(\omega_k * x) \quad (8)$$

$$(f \cdot g)(t) \triangleq \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau \quad (9)$$

$$\partial(i, j) = \text{optim}_{i \leq l \leq i+R-1, j \leq k \leq j+R-1} \{I(l, k)\} \quad (10)$$

Algorithm 1 Fingerprint authentication algorithm using the Gabor filter technique

Input: The fingerprint of the electorate.
Output: Fingerprint feature template file.
Begin:

Step 1 Capture the image of the fingerprint,
 $f(x, y) = (R, G, B)$.

Step 2 Convert the captured image from color to 8-bit grayscale $f_q(x, y) = (R + G + B)/3$.

Step 3 Resize the grayscale image to 258×288 pixels to obtain $f_{ar}(x, y)$.

Step 4 Normalize the image to obtain $f_{rgn}(x, y)$ using;
 $\frac{1}{a*b} \sum_{x=0}^{a-1} \sum_{y=0}^{b-1} f(x, y)$ such that;
 $a * b$ dimension of the image; 256×288 .

Step 5 Enhance $f_{rgn}(x, y)$ through histogram equalization;
 $f_{eq}(x, y) = \text{floor} \left((L - 1) \sum_{n=0}^{f_{x,y}} P_d \right)$
to obtain an equalized image, $f_{eq}(x, y)$, such that;
 $P_d = \frac{\text{number of density, } d \text{ pixels}}{\text{total number of pixels}}$
 $d = 0, 1, \dots, L - 1$.

Step 6 Eliminate the edges and noisy parts of the image through segmentation, using;
 $R_{xx} = \sum \sum_{(x,y)} R_x^2(x, y)$
 $R_{yy} = \sum \sum_{(x,y)} R_y^2(x, y)$
 $R_{xy} = \sum \sum_{(x,y)} R_x(x, y) \cdot R_y(x, y)$
 $\Theta(x, y) = \frac{1}{2} \tan^{-1} \left(\frac{2R_{xy}}{R_{xx} - R_{yy}} \right)$.

Step 7 Smooth and extract features of the image using a Gabor filter;
 $h(x) = e^{-\frac{1}{2} \left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2} \right)} \cdot \cos(2\pi f x_\theta)$.

Step 7 Smooth and extract features of the image using a Gabor filter;
 $h(x) = e^{-\frac{1}{2} \left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2} \right)} \cdot \cos(2\pi f x_A)$.

Step 8 Generate feature template using;

$$\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} = k \cdot \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & k^{-1} \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} \Delta_x \\ \Delta_y \\ \alpha \end{pmatrix}.$$

End

where,

 x is the input image ω_k is the convolutional filter of the k^{th} feature map $*$ is the convolutional operator f and g There are two integrable parameters from the input layer t is an input variable or pixel value ∂ is the output layer $\partial_{(i,j)}$ is the output matrix of i^{th} row versus j^{th} column R is the filter size

The rectified linear unit (ReLU) function is applied to the hidden layer as an activation function to enhance the training module. This convolution operation is achieved through max pooling and the application of ReLU functions to improve training operations. All the mapping images from the input features to the larger feature mapping are fully connected within the CNN model for the actual image prediction and matching. The electorate's facial image and the fingerprint

Algorithm 2 Facial image authentication using the Hough transform method

Input: Captured facial image of the electorate
Output: Facial feature template file.
Begin:

Step 1 Capture an image of the facial under near-infrared illumination.

Step 2 Convert the acquired image to 8-bit grayscale

Step 3 Segmentation of a grayscale image using $f_q(x, y) = (R + G + B)/3$.

$$\varepsilon(u, \Gamma) = \int_{\Omega - \Gamma} |u - u_0|^2 dx + \beta |\Gamma| + \nu \int_{\Omega - \Gamma} |\nabla u|^2 dx$$

Step 4 Localization of image using Hough transform;
 $(x - a)^2 + (y - b)^2 = r^2$
Such that;

$$\begin{cases} x_1 = r_1 \cos \theta + a_1 \\ y_1 = r_1 \sin \theta + b_1 \end{cases}$$

Step 5 Using a Gabor filter, extract a feature from the localized image;
 $h_{\{Re, I\} m} = \text{sgn}\{Re, I\} m \int_{\rho} \int_{\phi} I(\rho, \phi) e^{-i w(\theta_0 - \phi)} e^{-(r_0 - \rho)^2 / \alpha^2} e^{-(\theta_0 - \phi)^2 / \beta^2} \rho d\rho d\phi$

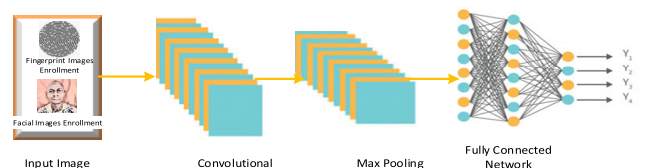
Step 6 For enrollment, store extracted features in a template mat file.

Step 7 For authentication;
i. Compare the newly captured mat file, A, with the stored mat file, B, using hamming distance;

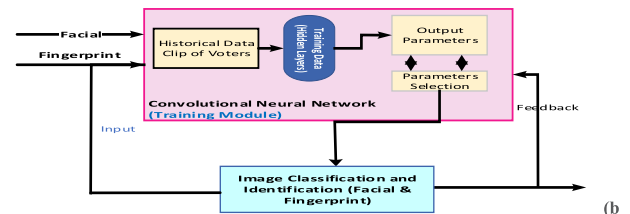
$$HD \text{ score} = \frac{\|(codeA \times codeB) \cap (maskA \cap maskB)\|}{\|maskA \cap maskB\|}$$

ii. if score > threshold;
authenticate;
else; reject.

End



(a). Overview of the DCNN model for image enrolment.



(b). Block module for multimodal biometric authentication system.

FIGURE 11. Deep convolutional neural network architecture for image processing.

enrolment template captured during the registration phase were optimized using histogram equalization, Hough transform, and Gabor filter for fast recognition and verification. These historical image clips were subjected to deep training

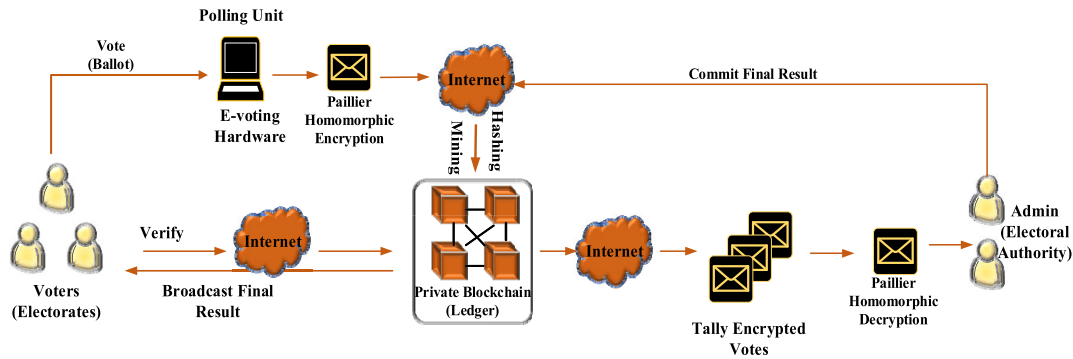


FIGURE 12. Secure decentralized system architecture.

for image recognition and matching using a CNN algorithm during the voting phase. Then, the system response in image verification and matching was excellent with low latency. The DCNN model overview of the bimodal biometric image processing is presented in Figure 11.

C. ETHEREUM BLOCKCHAIN TECHNOLOGY FOR SECURING E-VOTING SYSTEM

The blockchain-based decentralized database is designed for election registration, balloting, and election results. This blockchain is developed using Ethereum as the smart contract and the Solidity programming language, with enhanced performance infrastructure. Every transaction is executed at a faster rate, with a transaction per second (TPS) level. The Ethereum smart contract utilizes delegated proof of stake (DPoS) as a consensus algorithm to validate every transaction in the network, and the amount of energy needed for each transaction is validated. The Ethereum testnets use migrations.sol and migration.js as a pre-written file manager and smart contract project upgrading library, and for the smart contract script deployment. An ether.js and web3.js were configured to manage network information deployed on the decentralized applications, which connect Ethereum to the networks and recognize other related parameters.

The mining of each transaction (record) is implemented using a Structured Query Language (SQL). This mining technique is essential to combine a new record with the decentralized ledger database of the previous or existing one in the chain participant, and for validation. Homomorphic encryption was used to enhance the encryption of each transaction from the registration cycle to the polling result records. The additive and multiplicative homomorphic properties of the addition operations with the keys generated are used to store information. One of the two keys can be used for decryption. An overview of the secure, decentralized system-based architecture is shown in Figure 12.

The blocks are created for the set of e-voting record transactions in the Ethereum blockchain using a cryptographic hash algorithm, which serves as a foundational structure for the preceding blocks. This secure block makes the vote transaction immutable, tamper-resistant, and provides adequate security to the complete blockchain network. The block

Algorithm 3 Creation of a hash block for a secure, smart, verifiable e-voting system

Begin:

Error index κ_j ; Block $[\kappa_j]$; Ethereum blockchain Public address;
Total number of blocks created in Ethereum
Variable for κ_j of new block; Ethereum balance Cost threshold

Input: Block error index κ_j details.

Output: Block $[\kappa_j]$.

For $(1 \leq \kappa_j \leq m)$ Do: when
If $\kappa_j < m$, then:
Increment κ : $\xi \leftarrow \kappa + 1$;
Let E_{tb} for error index x
If $b_E \geq T_c$ then:
Create new block: Set Block $[\kappa_1]$.
Subtract cost: Let $E_{tb} - E_{tb} - T_c$.
Increment κ_y : $\kappa_j \leftarrow \kappa_j + 1$.
If $\kappa_j > m$, then:
End loop
End for

End

creation algorithm for the e-voting system is presented in Algorithm 3, and smart contract execution for the e-voting system is presented in Algorithm 4. As the number of voters in blockchain increases, the error index κ_j and the uniformity μ persistent level may also increase, which can be considered in the PoS consensus algorithm for blockchain scalability.

However, the execution of the smart contract is initialized by forwarding the transaction T_f that contained the electorate information, the smart contract with a unique identifier code was assigned. The transaction state, state variables S_v Present state of smart contract p_s state variable modification functions S_f and the number of n times (nth) inverse operation I_{or} are initialized to be executed in the algorithm.

The availability of the gas fee is used to check the transaction forwarded for the execution of the agreement on the smart contract platform. This sufficient gas fee on the smart contract platform will ensure that any transaction computation is successful, the transaction is forwarded, and the decline smart contract is updated by reverting the execution of the state changes. This smart contract execution algorithm implements transaction codes in the blockchain architec-

Algorithm 4 Execution of smart contract procedure for a secure, smart, verifiable e-voting system

Begin: Transaction forwarded T_f ; Smart contract state: State variables S_v ; Present state of smart contract P_s ; Final state of the smart contract F_s ; state variable modification functions Output; Inverse operation times I_{ot} ; Result of contract R_c

Input: Transaction forwarded T_f

Output: State of variable transaction.

Check: Ensure the gas fee is available to execute the transaction contract code.
 $Set P_s = run_he_mart_contract_code(state, T_f)$.
 Evaluate if validation is complete, then.
 Update the present state of the smart contract $P_{s-} = S_f(P_{s-}, varn)$.
 Run the inverse operation in nth times $T_f = inv(I_{ot})$
 For each smart contract transaction, the result in S_c
 Check, if possible, the contract state r
 $r \in S_c$ and $r = F_{S_c}$
 If true:
 Update smart contract state: $Set R_c = f(T_f, r)$
 Else
 Reset state: $Set R_c = S_o$
 Return R_c
 End if
 End else
 End for
 End

ture, which helps prevent unauthorized third-party access and malicious actors that can compromise the security mechanism. The smart contract algorithm ensures the validity of the final state, and the contract is immutable or open for modification.

1) ELECTORATE AUTHORITY

The Electoral Authority is an administrator who oversees the entire electoral process. The electoral authority first interacts with the e-voting system by enrolling the electorate, after which the electoral campaigns can be managed (either started or stopped). At the end of the electoral process, the electoral authority can proceed to tally the votes for the election, after which it can publish the results of the electoral campaign by sending the results to the email addresses of registered voters.

2) ELECTORATE

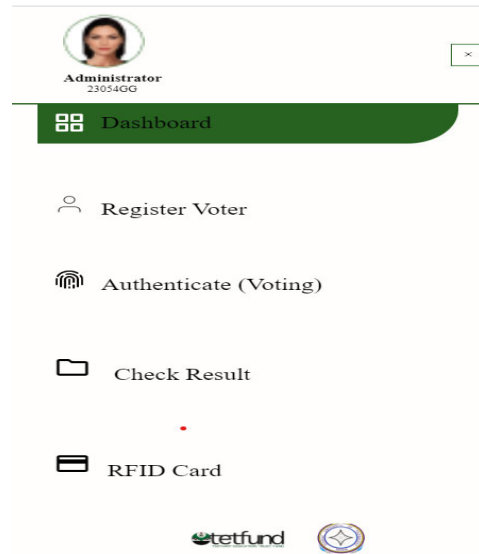
The electorate refers to people who have met the voting age requirement. They interact with the system through biometrics and a biodata form for enrollment. They are authenticated through a biometric system to cast their votes. At the end of the electoral process, the e-voting system allows the electorate to audit the electoral process through the publicly accessible ledger of the blockchain.

IV. RESULTS AND DISCUSSIONS

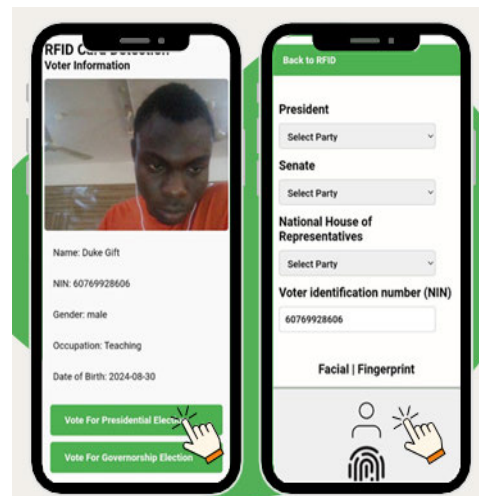
The experimental activities of a secure, smart, and verifiable decentralized e-voting system begin with voter registration, secure account creation, biodata form, and biometric information enrollment, as depicted in Figures 13(a) and 13(b). The biodata, facial, and fingerprint image capturing is



(a) Login personal account creation graphic user interface (GUI).

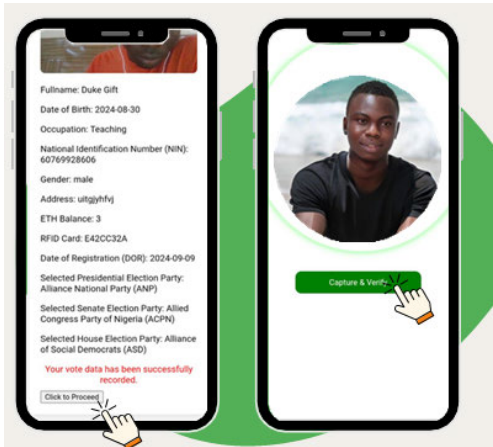


(b) Voter registration dashboard GUI.

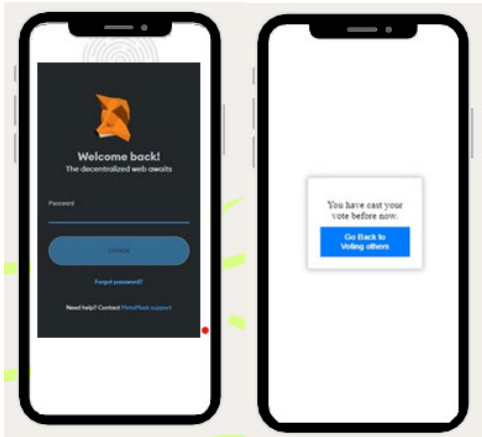


(c) Bimodal biometric user enrolment GUI.

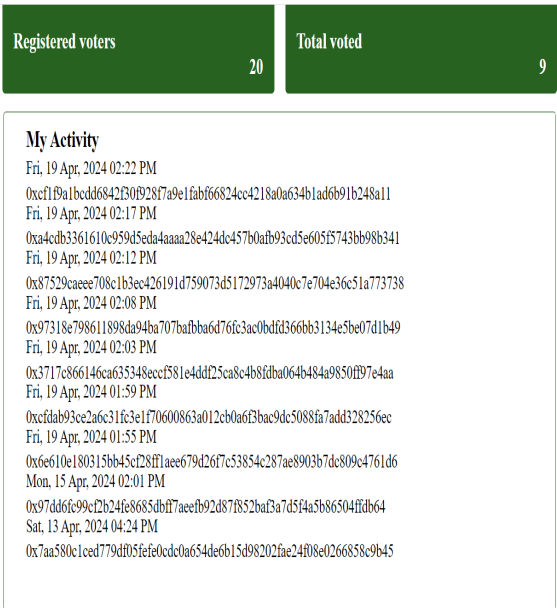
FIGURE 13. The GUI analysis of decentralized SSVEVS during the electioneering process includes a login interface for the administrator (electoral authority) and electorate, biometric capturing, an electorate verification interface, Metamask for securing voter information, an electorate records interface, and vote result analysis.



(d) Biometric user enrollment confirmation GUI.



(e) Metamask GUI.



(f) Voter registration record GUI.

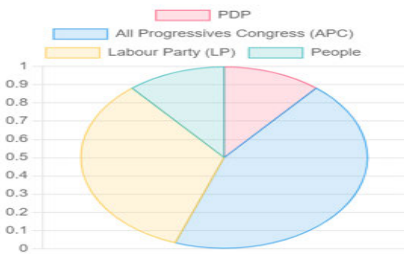
FIGURE 13. (Continued.) The GUI analysis of decentralized SSVEVS during the electioneering process includes a login interface for the administrator (electoral authority) and electorate, biometric capturing, an electorate verification interface, Metamask for securing voter information, an electorate records interface, and vote result analysis.

Election Result:

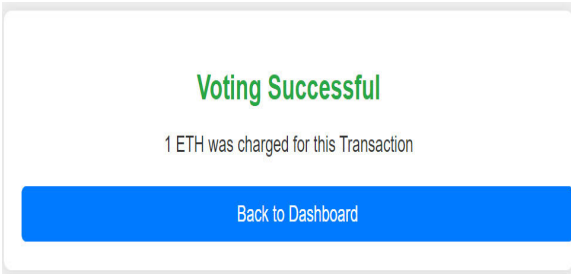
Election	Party	Count
President Election	PDP	1
Presidential Election	All Progressives Congress (APC)	4
Presidential Election	Labour Party (LP)	3
Presidential Election	People	1
Total	PDP	1
Total	All Progressives Congress (APC)	4
Total	Labour Party (LP)	3
Total	People	1

(g) Analysis of election results.

Pie Chart



(h) Graphical analysis of the election results.



(i) Successful voting GUI and charge for each transaction.

FIGURE 13. (Continued.) The GUI analysis of decentralized SSVEVS during the electioneering process includes a login interface for the administrator (electoral authority) and electorate, biometric capturing, an electorate verification interface, Metamask for securing voter information, an electorate records interface, and vote result analysis.

encrypted using the PHC public key pk generated during the pre-voting phase at the polling unit, as shown in 13(c) and 13(d). This encrypted information is verified using private key fk during the voting phase, and at the final stage (vote counting) for auditability. All the registration information and the votes cast are encrypted using the Homomorphic encryption algorithm. The data, the voter's Blockchain ID, and the previous block are created and mined into the private blockchain ledger for the electorate's use through a MetaMask platform, as shown in Figure 13(e).

The message received contains the cipher of the ballots and the PIN registration link. This message link is used to verify the uniqueness of the cipher and compare the cipher value of the ballot in the email with the registered value on the blockchain-based public ledger. This cipher or unique PIN is for the voting process without compromising the privacy of the electorate. Hence, it prevents multiple voting due to the uniqueness of the PIN generated and hashed. This process is sustained until the admin (electoral authority) decides to tally

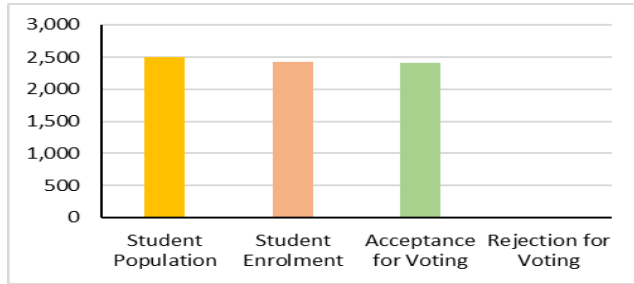


FIGURE 14. Analysis of students' enrollment for the election.

the encrypted ballots to obtain a single final encrypted sum value, which is decrypted homomorphically.

Then, the electoral authority commences the voter enrolment process through a wireless RFID tag or biometric user verification. This RFID card/tag serves as the initial level of authentication and identification during the voting phase. It is considered valid if the electorate's information matches, allowing the electorate to proceed with biometric image capture and verification. Alternatively, the electoral biometric identities (facial and fingerprint) captured can be used for the electorate verification during voting by selecting the party's logo and election type. The registered candidates and votes cast are encrypted (hashed) and added to the blocks of the blockchain network. The token is assigned to each electorate and charged a lower amount of gas fee for every transaction initiated accordingly. Otherwise, if the biometric image mismatches after three trials, the electorate will be denied the opportunity to proceed with the voting exercise. The secure voters' registered records, vote analysis, and successful page for each electorate are depicted in Figure 13.

A. TESTING AND VALIDATION OF SSV E-VOTING SYSTEM

The flexibility, robustness, and efficiency of the developed SSVEVS were tested and validated during the 2024 electioneering process of the Nigeria Association of Computer Engineering Students (NACOMES), Federal University of Technology, Minna. The Department comprises levels (100-500) with a total population of 2,500 students. The total number of students enrolled through the SSVEVS platform during the electioneering process is 2,423. The total number of candidate acceptances for voting during verification stages is 2,422 (99.96%), and only one candidate was rejected (0.04%) due to policy compromise. Otherwise, we achieved a 100% voting acceptance rate with 0.00% rejection rate. The graphical analysis is shown in Figure 14.

The time taken for each candidate's enrollment varies due to several factors, including biometric matching challenges such as facial and fingerprint blurs, network signal stability, and personal disturbances. The notable average time for candidates during the registration phase ranges from 125 to 240 seconds. During verification and voting stages, the average time ranged from 90 to 150 seconds. Therefore, the overall average processing time of the secure

TABLE 5. Result analysis of blockchain transaction.

CGS (B)	CGT (ms)	DRS (B)	DRT (ms)
82.28920182	3.235198058	175.6954745	100.1922758
81.58539445	3.82662708	172.5488684	100.7711419
86.57579995	3.763682638	170.892539	143.7090605
85.71546037	4.952976748	176.7374705	108.2686246
84.41077645	4.310570692	176.3938291	133.4306922
80.44476473	4.612636437	176.0691037	109.5396121
87.22432448	3.868658416	173.9071339	119.9189385
84.12443112	4.280840018	175.890501	105.4529058
88.83949617	4.135751015	176.5756164	118.5715635
84.58413064	4.780242538	178.3411618	122.2554693
88.34959902	5.015301737	172.2427407	135.5951668
86.50646864	4.711573656	172.7411769	146.8890459
83.62595246	4.227967731	176.9166234	125.0213717
85.1094683	3.366397117	176.9562713	145.0971192
89.40776388	5.153466058	173.8478512	99.01464416
84.24071391	4.802034726	177.4890188	122.2698725
82.35036759	5.00881042	177.5671357	121.5789208
84.25226317	4.781515119	171.2440345	108.930891
88.19642409	5.137781421	173.1252766	127.2121494
80.03250242	5.15847563	177.3029929	143.3343711
85.46882	3.123496716	174.4493348	103.588831
88.4346864	3.180903107	172.8963084	101.1403589
82.41144178	5.068985347	172.7070293	126.8990377
87.94386493	4.301151127	175.8786328	131.3807238
80.84511812	3.429885596	171.1784316	111.0288953
84.67154173	4.540471005	171.5078943	107.2097332
86.99563725	5.293412514	175.717887	145.4204718
85.01409801	4.63714864	174.8942071	108.5501555
82.5471793	3.820012478	172.4183681	105.0276676

voting system, considering registration, verification, and voting periods, is estimated to be 302.5 seconds per candidate.

B. EVALUATION OF BLOCKCHAIN TECHNOLOGY-BASED SECURE E-VOTING SYSTEM

The Ethereum blockchain is deployed on the smart contract platform using Solidity to improve the e-voting system. The smart contract uses the MetaMask Chrome extension cryptocurrency wallet and web3 library for interaction and connectivity on the Sepolia Testnet blockchain-based decentralized framework. This framework helps verify and validate every encrypted transaction (vote), which is mined and hashed into blocks for public records. Mining is the process by which nodes on the blockchain (miners) verify the validity of the transactions in the new block using a consensus protocol called smart contracts. The open-source, smart contract-based blockchain adopts the Proof-of-Stake (PoS) consensus protocol, which ensures that no node in the e-voting system has excessive influence over others. The consensus process begins by adding a nonce value to the new block. The resulting combination is processed through the SHA256 hashing function to produce a digest value whose prefix is compared with the required hash prefix.

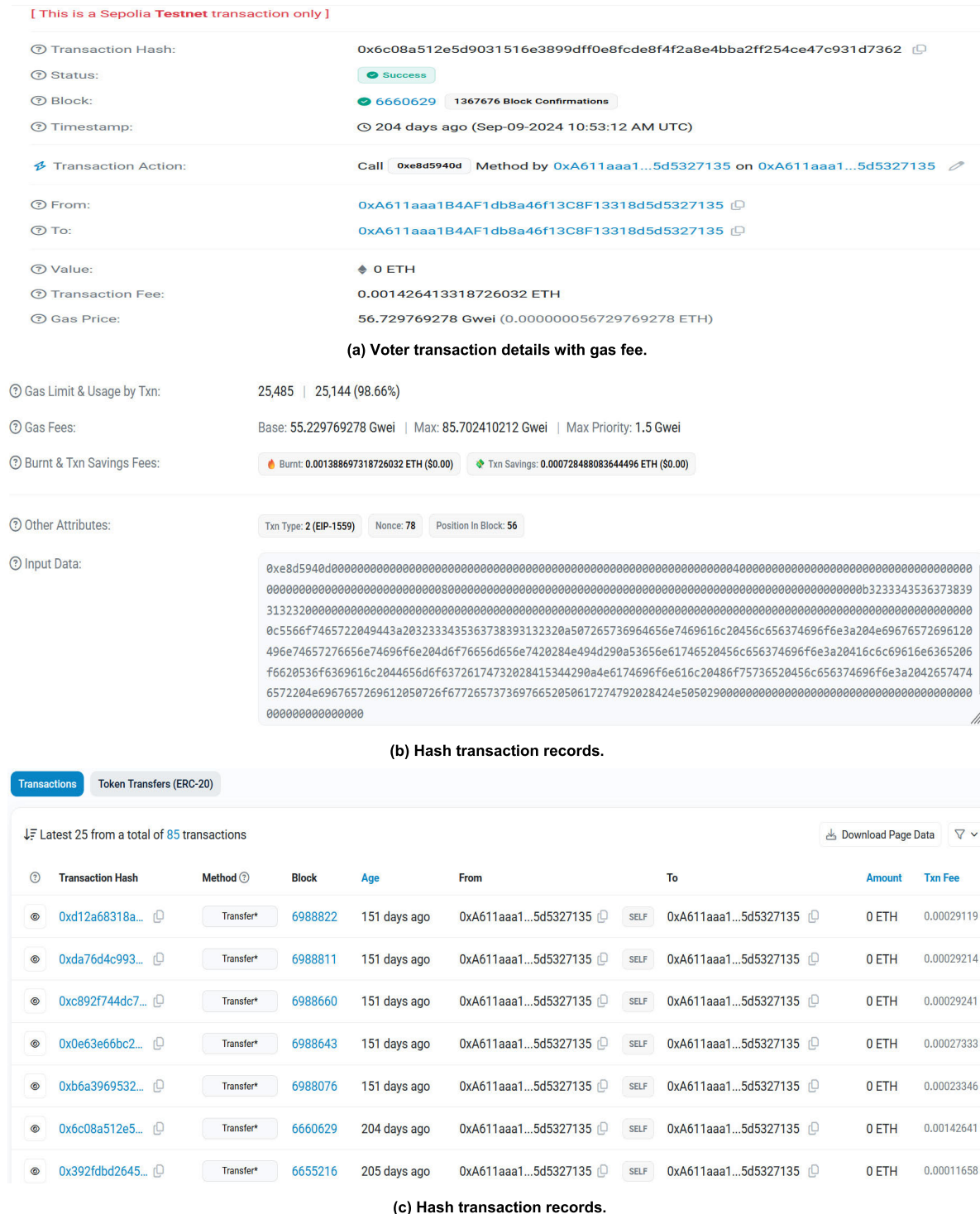


FIGURE 15. The GUI of the transaction and the hashed information during voting.

If the digest matches the required prefix, the new block is added to the blockchain; otherwise, the nonce value is changed repeatedly until the hash prefix condition is met. The graphical user interface displays the hashed information, Sepolia Testnet transactions, and analysis of each electorate transaction, including transfer fees in the SSVEVS, as shown in Figure 15. Each transaction's details include code, block

status, timestamp, source and destination hashes, Ethereum value, transaction fee, and gas fee, all detailed in the Sepolia Testnet transaction GUI.

The certificate size generated (bytes), data retrieval size (bytes), and time (milliseconds) for each transaction on the Sepolia Testnet-based Blockchain platform during the voting process are presented in Table 5.

V. CONCLUSION

This research presents a secure, smart, and verifiable electronic voting system (SSVEVS) to ensure trust in the general process. The system model combines a multimodal authentication system (RFID, facial, and fingerprint) for verification, and offers adequate security (authentication and authorization) as a measure for an electorate's eligibility. The Ethereum blockchain enhances the security mechanism with assurance of vote tamper-resistance, transparency, trust, and verifiability. However, cross-correlation of the security model gives good accuracy in image matching, transaction records, and vote verifiability. However, the design of SSVEVS was tested and validated during the NACOMES-FUTMinna 2024 election. The true acceptance rate of the electorate during verification and votes cast is 100% accurate. A false image template with a veil, recognized by 2,422 (99.96%) of the population, and only one candidate was rejected (0.04%) due to policy compromise. The system response time was observed for each candidate during the registration phase, consumed between 125 and 240 seconds, and vote casting took 90 to 150 seconds. Therefore, the average processing time of the secure voting system is estimated to be 302.5 seconds per candidate. The transaction of each record on the Sepolina Testnet Blockchain platform is noted with a maximum transaction fee of 0.00029199ETH. The certificate size generated ranges between 82.55 and 86.99 bytes, and the time taken for the certificate is between 3.2351 and 5.2934 milliseconds. And ensure trust, privacy, transparency, and immutability. Future work will focus on comparative analysis and challenges to the best suitability of cryptographic methods with blockchain technology for securing the e-voting system.

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